Hales, Stephen (1677–1761) English botanist and clergyman whose pioneering research on plant and animal physiology was described in his *Vegetable Staticks*. The first to note the upward flow of sap and measure vapor emission in plants, he also measured blood pressure and blood output from the heart. His inventions included an artificial ventilator and pneumatic trough.

Halley, Edmond (1656–1742) English astronomer and mathematician who calculated the orbit of the eponymous Halley's Comet and its subsequent 1758 date of return to Earth. Later Astronomer Royal, Halley published influential papers on magnetic variation, the trade winds and monsoons. He was also responsible for the publication of Isaac Newton's *Principia*.

Harrison, John (1693–1776) English carpenter and clockmaker who invented the marine chronometer, which enabled sailors to establish their position at sea. Harrison designed and built four chronometers in response to a £20,000 government prize offered in 1714 for a way of accurately finding longitude at sea. Despite the great accuracy of his chronometers, Harrison was not paid in full until 1773.

Harvey, William, English physician (1578–1657) See p.103

Hawking, Stephen, British theoretical physicist (1942–) See p.305

Heisenberg, Werner, German physicist (1901–76) See p.259

Henry, Joseph (1797–1878) American physicist who discovered the phenomenon of self-inductance—a defining principle of electronic circuitry. His many contributions included constructing the first electromagnetic motor, developing the telegraph with Samuel Morse, and introducing an early weather forecasting system.

Herschel, Caroline (1750–1848) German-born British astronomer who had a long collaboration with her brother, William Herschel. Planning to be an opera singer, she moved to her brother's house in England at age 22, and is noted for discovering three nebulae and eight comets, as well as for completing their star catalog.

Herschel, William (1738–1822) German-born British astronomer noted for discovering Uranus in 1781. Originally a music teacher, Herschel took up astronomy and specialized in making very large telescopes. He developed a theory of nebulae and star evolution, observed and cataloged many stars, and showed that the Solar System moves through space.

Hertz, Heinrich, German physicist (1857–94) See p.224

Hertzsprung, Ejnar (1873–1967) Danish astronomer best known for his 1913 Hertzsprung–Russell diagram, a star classification system still used today. The diagram, developed with Henry Norris Russell, plotted the brightness of stars against their spectral types. He also researched open star clusters and variable stars, and developed a method for positioning double-stars.

Hevelius, Johannes (1611–87) Polish astronomer and early lunar topographer, best known for his detailed map of the Moon's surface. A city councillor of Gdansk, Hevelius built an observatory on top of his house to investigate the night sky. He cataloged over 1,500 stars, discovered several constellations, and named many lunar features.

Higgs, Peter, British physicist (1929–) See p.348

Hipparchus (*c.*170–*c.*120 BCE) Greek astronomer and mathematician often considered the founder of trigonometry. Hipparchus's contributions to astronomy include a study of solar eclipses, discovery of the precession of the equinoxes, and a description of the Sun and Moon's orbits and their distances from Earth.

Hippocrates (*c.*460–*c.*377 BCE) Greek physician widely regarded as the father of medicine. A medical pragmatist, Hippocrates based his practice on studies of the body and the symptoms and treatments of illness. He was the first to describe many diseases and coin terms such as "acute," "chronic," and "relapse." Hippocrates' code of ethics for his medical students is today known as the Hippocratic oath sworn by all doctors.

Hodgkin, Dorothy, British chemist (1910–94) See p.275

Hooke, Robert (1635–1703) English inventor and natural philosopher who, after being Robert Boyle's assistant, became Curator of Experiments at the newly established Royal Society in London. He worked on theoretical astronomy, as well as inventing a compound (two-lens) microscope to study microscopic life—producing the Royal Society's first publication: *Micrographia*. He was also the first to record biological cells. Given the range of his scientific contribution, he is widely hailed as England's Leonardo.

Hopper, Grace (1906–92) American mathematician and pioneer of computer programming and technology. A rear admiral in the US Navy, Hopper was one of the first programmers of the Harvard Mark I and helped develop UNIVAC I, the first commercial electronic computer. She also contributed to the COBOL computer language, and introduced the term "bug." US Navy's missile-destroyer, the *Hopper*, is named after her.

Hoyle, Fred, British mathematician and astronomer (1915–2001) See p.280

Hubble, Edwin (1889–1953) American astronomer considered the founder of extragalactic astronomy for discovering that the Universe is expanding. While working at the Mount Wilson Observatory, California, Hubble established that previously thought nebulae of the Milky Way were in fact different galaxies receding away from our own. The rate at which the Universe is expanding is known as the Hubble constant.

Humboldt, Alexander von (1769–1859) German naturalist, explorer, and pioneer of biogeography best known for investigating the geography and flora and fauna of Latin America with French botanist Aimé Bonpland. An enthusiastic popularizer of science, Humbolt spent 25 years writing *Cosmos*, an account of the structure of the Universe—four volumes of which were published during his lifetime.

Hutton, James, British geologist (1726–97) See p.157

Huxley, Thomas Henry (1825–95) English biologist, surgeon, and champion of Darwinism. His studies on comparative anatomy led him to conclude that birds evolved from dinosaurs. He debated evolutionary theory against Samuel Wilberforce in 1860, earning him the nickname "Darwin's bulldog." Huxley also declared himself agnostic—a term he coined.

Huygens, Christiaan (1629–95) Dutch physicist, mathematician, and astronomer known for the Huygens–Fresnel principle, which states that light is made up of waves. Huygens discovered the rings of Saturn and its fourth moon Titan, and also invented the pendulum clock and other time-keeping innovations.

Ibn Sina (Avicenna), Persian physician (980–1037) See p.50

Ingenhousz, Jan, Dutch physician (1730–99) See p.155

Isidore of Seville, Saint, Spanish theologian (*c.*560–636 CE) See p.42

Jeans, James (1877–1946) English physicist, mathematician, and astronomer. A great popularizer of astronomy, Jeans investigated spiral nebulae, multiple star systems, and giant and dwarf stars. He was also the first to hypothesize a continuous creation of matter throughout the Universe. Among his best-known books is the 1929 *The Universe Around Us*.

Jenner, Edward (1749–1823) English physician who developed a vaccine for smallpox. Learning from dairymaids, Jenner observed that a person infected with the cowpox disease would not succumb to the deadly smallpox virus. Within five years, his cowpox inoculation was in widespread use. Smallpox was declared eradicated in 1980.

Joule, James Prescott (1818–89) English physicist who provided the foundation for the theory of conservation of energy, which states that energy can change form, but cannot be created or destroyed. Joule showed that heat is energy and helped establish the mechanical equivalent of heat.

Kamerlingh Onnes, Heike (1853–1926) Dutch physicist awarded the 1913 Nobel Prize in Physics for his research on low-temperature physics and for discovering liquid helium. His work in cryogenics led him to discover superconductivity.

Kant, Immanuel (1724–1804) German philosopher whose theories on knowledge, ethics, and esthetics profoundly influenced subsequent philosophical thought. Kant attempted to reconcile the theories of rationalism (we know only what our minds can construct) and empiricism (we know only what our senses reveal) by asking "what can we know?"

Kekulé, Friedrich August (1829–96) German chemist and founder of structural theory in organic chemistry. A professor at Ghent and Bonn universities, Kekulé showed that carbon atoms can link together to form chains, which led to his later discovery of benzene's six-carbon cyclical structure.

Kepler, Johannes, German astronomer (1571–1630) See p.95

Khayyam, Omar, Persian mathematician and astronomer (1048–1131) See p.53

Koch, Robert (1843–1910) German physician awarded the 1905 Nobel Prize in Physiology or Medicine for isolating the tuberculosis bacillus. Considered one of the founders of microbiology and bacteriology, Koch also discovered the bacteria responsible for anthrax and cholera. His postulates establish four criteria for investigating the relationship between a causative microbe and a disease.

Krebs, Hans (1900–81) German–British physician and biochemist who discovered the citric acid cycle in living organisms. The discovery of this metabolic cycle, also known as the Krebs cycle, won him and Fritz Lipmann the 1953 Nobel Prize in Physiology or Medicine. Krebs also discovered the urea cycle, during which mammals convert ammonia into urea.

Lamarck, Jean-Baptiste, French biologist (1744–1829) See p.169

Laplace, Pierre-Simon (1749–1827) French astronomer and mathematician noted for his research into the stability of the Solar System and often called "the French Newton." Using calculus, Laplace reformed astronomical mathematics in his five-volume *Celestial Mechanics*, and introduced determinism into Newtonianism. Several operators and transforms are named after him.

Laue, Max von (1879–1960) German physicist awarded the 1914 Nobel Prize in Physics for studying the diffraction of X-rays in crystals. This proved important for X-ray crystallography, solid-state physics, and modern electronics. Director of the Max Planck Institute and the Institute for Theoretical Physics, Laue also researched into superconductivity, quantum theory, and optics.

Lavoisier, Antoine Laurent, French chemist (1743–94) See p.160

Lawrence, Ernest O. (1901–58) American physicist awarded the 1939 Nobel Prize in Physics for inventing the cyclotron, which accelerates particles to study subatomic interactions. Lawrence used his cyclotron to produce radioactive iodine, phosphorus, and other isotopes for medical use. A professor at University of California, Berkeley, Lawrence later contributed to the Manhattan Project. The element lawrencium is named after him.

Leakey, Louis, British archaeologist and anthropologist (1903–72) See p.289

Leakey, Mary, British archaeologist and paleontologist (1913–96) See p.308

Leavitt, Henrietta Swan (1868–1921) American astronomer who discovered the relationship between the brightness and time span of Cepheid variable stars. Leavitt worked at the Harvard College Observatory, where she examined the luminosity of stars from photographic plates, and observed that Cepheid variable stars showed a regular pattern of brightness. Her work proved crucial for measuring the distance between Earth and other galaxies.

Lee, Tsung-Dao (1926–) Chinese-born American physicist and co-winner of the 1957 Nobel Prize in Physics for discovering violations of the law of parity conservation, which led to important developments in particle physics. Lee created a solvable model of quantum field theory, called the Lee model, and helped study the violations of time-reversal invariance.